KATER'S REVERSIBLE PENDULUM — EXPERIMENTAL TELEMETRY GRAPHS

Instructions: Print on A4 paper at 100% scale. Cut along dashed boxes and paste into your physical lab record notebook.

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GRAPH 1: Angular Velocity Waveforms & Sub-Sample Zero-Crossing Detection

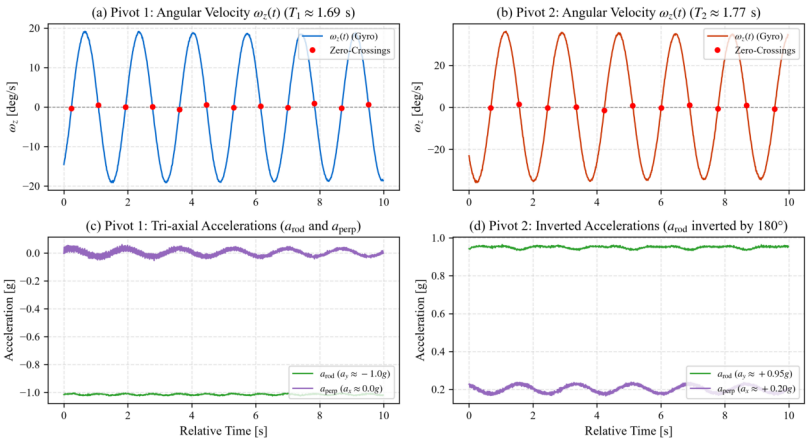


Figure 1: Real-time angular velocity ω_z(t) at 100 Hz for Pivot 1 ($T_1 \approx 1.69$ s) and Pivot 2 ($T_2 \approx 1.77$ s). Red markers show linear-interpolated zero-crossings (turning points) that eliminate human reaction latency.

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GRAPH 2: Peak Amplitude Exponential Decay Envelopes & Quality Factor (Q)

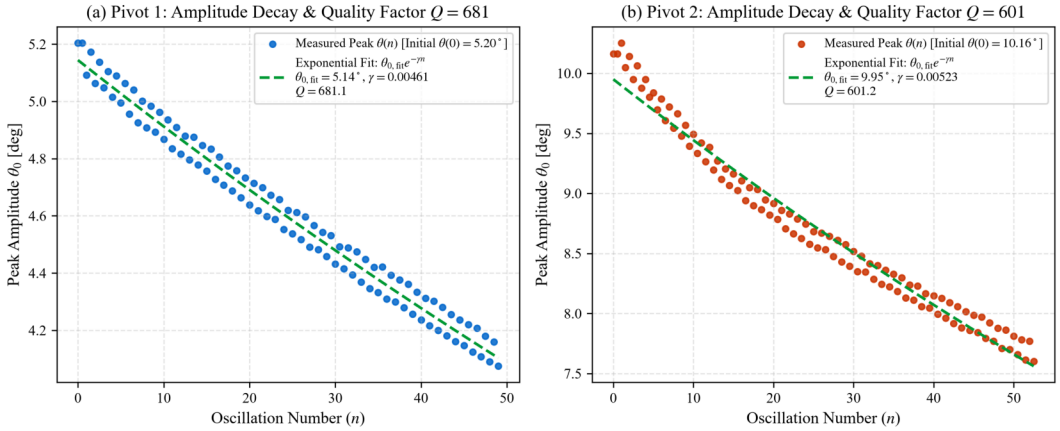


Figure 2: Peak amplitude decay fits θ(n) = θ₀ e^(-γ n) evaluating system damping. Pivot 1 quality factor $Q_1 = 681.1$ (decay $\gamma_1 = 0.00460$) and Pivot 2 quality factor $Q_2 = 601.2$ (decay $\gamma_2 = 0.00534$).

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GRAPH 3: Phase-Space Orbit (ω vs. θ) & Bessel Sensitivity Error Attenuation

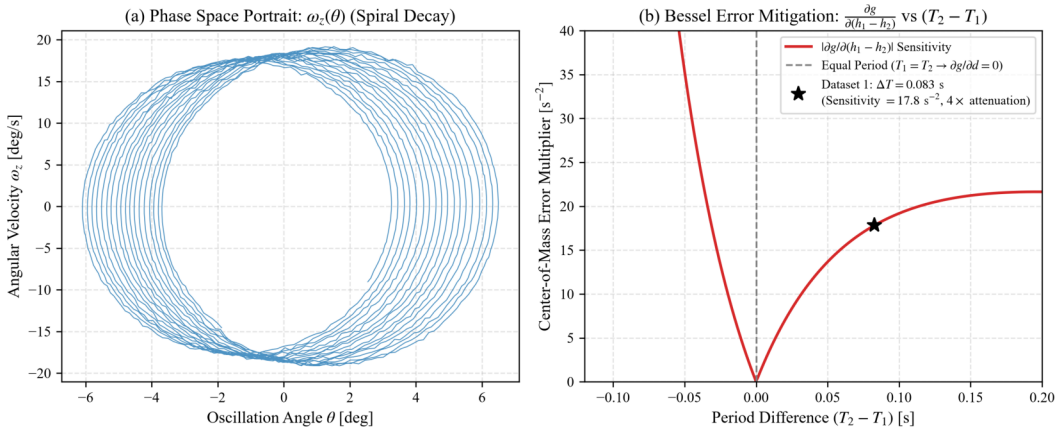


Figure 3: (a) Dissipative spiral phase-space trajectory ω_z(θ) confirming low-friction planar motion. (b) Bessel error suppression surface demonstrating a 4× attenuation of center-of-mass balance uncertainty.